

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (EE) Examination

Nov 2013

EE- 215 Linear Electronics

Date: 26.11.2013, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Describe the working of centre-tap full-wave crystal diode rectifier. What is the PIV of each diode? Mention disadvantages of this circuit. [05]
- (b) What do you understand by intrinsic and extrinsic semiconductors ? [02]
- Q - 2 (a) Discuss the action of a zener diode as a voltage stabilizer. [06]
- (b) Define α . Derive the relation between β and α . [04]
- (c) For a transistor, $\beta = 50$ and voltage drop across $1K\Omega$ which is connected in the collector circuit is 1 volt. Find the base current for common emitter connection. [04]

OR

- Q - 2 (a) Draw and explain V-I characteristics of a pn junction. [06]
- (b) What is zener diode? What is the difference between zener diode and ordinary diode? [04]
- (c) Give the comparison of the three basic rectifier circuits. Which is the best circuit ? [04]
- Q - 3 (a) Draw and explain input and output characteristics of CE connection. [06]
- (b) In common emitter connection of transistor $V_{cc} = 12.5V$, $R_c = 2.5K\Omega$, draw the d.c. load line. what will be the Q point if zero signal collector current is 1mA [04]
- (c) Which transistor connection is commonly used and why? [04]

OR

- Q - 3 (a) Establish the following relations: [06]
- (i) $I_E = (\beta+1) I_B + (\beta+1) I_{CBO}$ (ii) $I_c(1 - \alpha) = \alpha I_B + I_{CBO}$
- (b) Explain i) capacitor filter ii) π -filter. [04]

(c) Fill in the blanks by inserting appropriate words [04]

- i) An ideal value of stability factor is _____.
- ii) If the value of α is 0.9, the value of β is _____.
- iii) The maximum efficiency of a half-wave rectifier is _____.
- iv) The dimension of h_{ie} parameter is _____.

SECTION – II

Q - 4 (a) Derive the general formula for the i) Input Impedance (ii) Current gain (iii) Voltage gain in terms of h-parameters and load. [07]

Q - 5 (a) Describe the feedback resistor method of transistor biasing. [06]

(b) Give the difference between JFET and BJT. [04]

(c) Define the JFET parameters and establish the relationship between them. [04]

OR

Q - 5 (a) Explain potential divider method of JFET biasing. [06]

(b) Discuss the cut off, saturation and active regions of the output characteristics of CE transistor circuit. [04]

(c) The following quantities are measured in a CE amplifier circuit: [04]

(i) With output a.c short-circuited (i.e. $V_{ce} = 0$)

$$I_b = 10 \mu A, I_c = 1 mA, V_{be} = 10 mV$$

(ii) With input a.c open-circuited (i.e. $I_b = 0$)

$$V_{be} = 0.65 mV, I_c = 60 \mu A, V_{ce} = 1 V.$$

Determine all the four h parameters.

Q - 6 (a) Explain the construction and working of MOSFET. [06]

(b) What is stability factor? Derive the equation of stability factor [04]

(c) It is desired to set $R_1 = 10 K\Omega$, $R_2 = 5 K\Omega$, $R_c = 1 K\Omega$, $R_E = 2 K\Omega$, $V_{CC} = 15V$ by biasing a silicon transistor with voltage divider bias method. Draw the d.c. load line and determine the operating point. [04]

OR

Q - 6 (a) Explain following terms of a JFET circuit. [06]

- i) Shorted-gate drain current
- ii) Pinch off voltage
- iii) Gate-source cut off voltage

(b) How will you determine the drain characteristics of JFET ? What do they indicate ? [04]

(c) Give the difference between MOSFET and JFET. [04]
